

Lab Assignment 6: Logistic Regression on MNIST Dataset

Setup Note:

Full MNIST contains 70,000 images, which can cause slow training and test-time latency. Use a subset of 10,000 training images and 2,000 test images for all tasks so code runs in seconds.

You may use the following code snippet to start:

```
from sklearn.datasets import fetch_openml

from sklearn.model_selection import train_test_split

# Load and scale to [0, 1]
X, y = fetch_openml("mnist_784", version=1, return_X_y=True, as_frame=False)
X = X / 255.0

# Subsample for 2-hour lab runtime
X_train, X_test, y_train, y_test = train_test_split(
    X, y, train_size=10000, test_size=2000, random_state=42, stratify=y
)
```

TASK 1 Exploratory Visualization & Baseline Classifier

1.1 Sample Inspection: Reshape and plot a 2×5 grid showing one representative training image for each digit class (0 through 9) with its corresponding ground-truth label.

1.2 Baseline Classifier Fitting: [CONSULT DOCS]

Look up the linear classification models in scikit-learn. Instantiate and fit a multinomial logistic regression model on your training subset.

Which optimization solver is used by default? Did the model converge within default iterations, or did you need to increase the iteration limit? Measure and report the training elapsed time and the overall accuracy on your test subset.

1.3 Error Diagnostics via Confusion Matrix: [CONSULT DOCS]

Explore the evaluation metrics module in scikit-learn to compute and display a normalized confusion matrix visualizer.

Question 1.3: Identify the two digit pairs exhibiting the highest mutual confusion rate. In your own words try to explain why it might be the case?

TASK 2 Resolution Scaling & Downsampling Trade-offs

Evaluate the resilience of linear models to spatial information loss by downsampling images to 14×14 and 7×7 .

2.1 Downsampling Pipeline: [CONSULT DOCS]

Downsample all images in both X_{train} and X_{test} from their original 28×28 dimensions down to:

Downsample all images in both train and test sets to:

Resolution A: 14×14 pixels

Resolution B: 7×7 pixels

(Hint: Use `skimage.transform.resize`. Because each sample in X_{train} and X_{test} is stored as a flattened 784-dimensional vector, reshape each image back to $(28, 28)$ before resizing, then flatten (`.ravel()`) it back. You can apply this across the dataset using a loop or list comprehension:)

Python Hint :

```
from skimage.transform import resize
```

```
# Example: downsample a single 1D image vector 'x' (shape: 784,) to 14x14
```

```
resized_sample = resize(x.reshape(28, 28), (14, 14), anti_aliasing=True).ravel()
```

```
# To process the full split:
```

```
# X_train_14 = np.array([resize(img.reshape(28, 28), (14, 14), anti_aliasing=True).ravel() for  
img in X_train])
```

2.2 Resolution Comparison Experiment: Train identical logistic regression models on each resolution.

Note down the test accuracy of each.

Question 2.3: When reducing the number of features by over 93% ($784 \rightarrow 49$), how much accuracy did the linear model lose? Why does a linear classifier remain relatively effective even when digits are blurred?

TASK 3 Interpretability & Spatial Weight Map Visualization

Unlike black-box architectures, multinomial logistic regression produces direct linear weights that map back to the input coordinate space.

3.1 Parameter Discovery: [CONSULT DOCS]

Check the documentation for your fitted logistic regression model. Identify the public attribute storing the learned coefficient matrix.

- What is the attribute name and its array shape? What do the dimensions correspond to?
- Reshape each class's weight vector back into the original 28×28 2D image coordinates.

3.2 Diverging Heatmap Visualization:

Plot all 10 weight vectors in a 2×5 subplot layout using a diverging colormap centered symmetrically at zero (e.g., set `vmin=-scale` and `vmax=scale`, where `scale=np.max(np.abs(weights))`) along with a colorbar.

Question 3.3 (Visualizing Decision Ambiguity):

Take the two most confused digits identified in Task 1.3. Compare their weight heatmaps side by side. How does spatial overlap or cancellation between their weights explain why the model mistakes one for the other?

TASK 4 Parametric vs. Non-Parametric Benchmark (k-NN)

4.1 Estimator Discovery: [CONSULT DOCS] Look up the k-Nearest Neighbors classification estimator in scikit-learn. Fit a 3-NN model on the 14×14 downsampled dataset (do not run on 28×28 to avoid long distance-computation delays).

4.2 Benchmark: Measure and compare the test accuracy and test-set prediction latency (inference time on the 2,000 test images) between Logistic Regression (14×14) and 3-NN (14×14).

Question 4.3: Express the test-time computational complexity of both algorithms in Big-O notation in terms of training samples N and feature count d . Why is Logistic Regression faster for real-time prediction? [Non Graded but to be attempted]